B.Tech. CSE VI CSS 338

EMBEDDED SYSTEMS LAB

Name: Samina Parveen

Roll Number: BTECH/10007/23

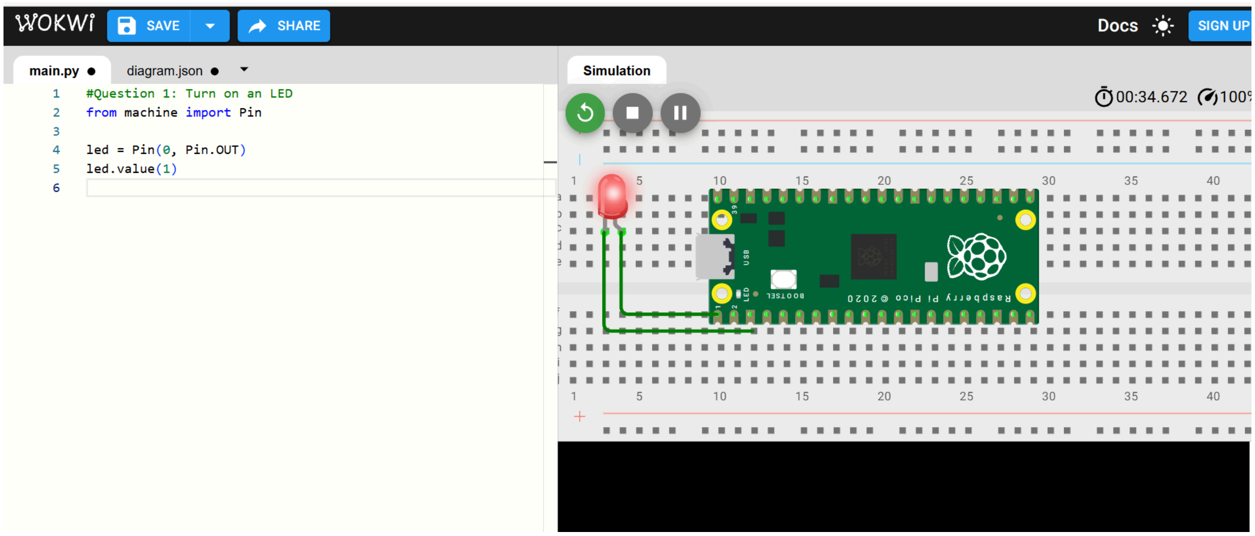
Assignment-1

Using Micro Python, simulate the following on a Raspberry Pi Pico:

1. **Turn on an LED**

   Code

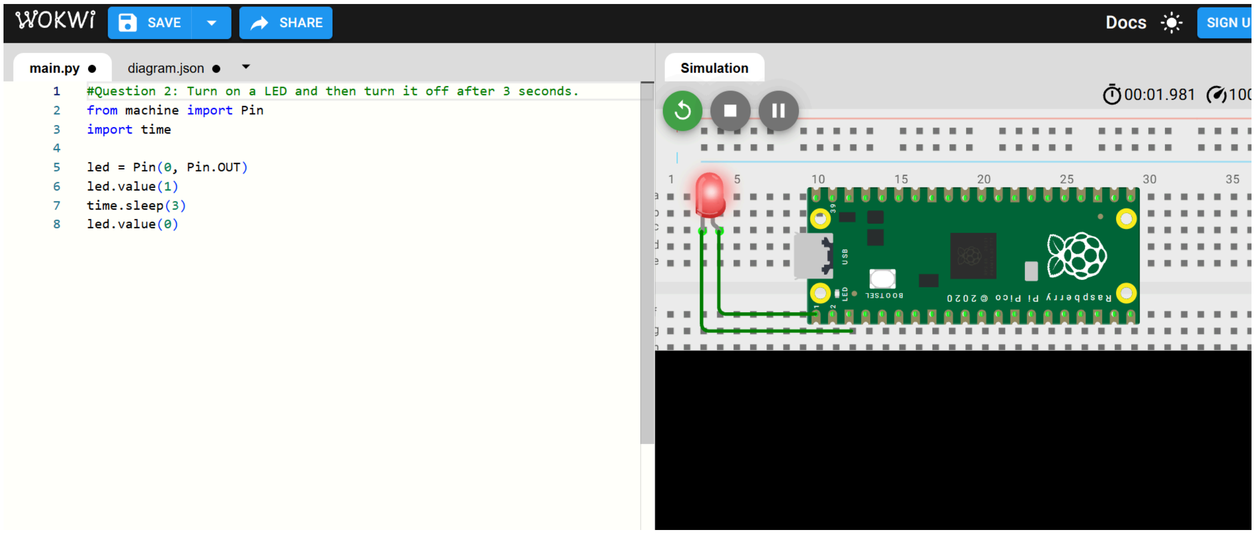
```
from machine import Pin
led = Pin(0, Pin.OUT)
led.value(1)
```

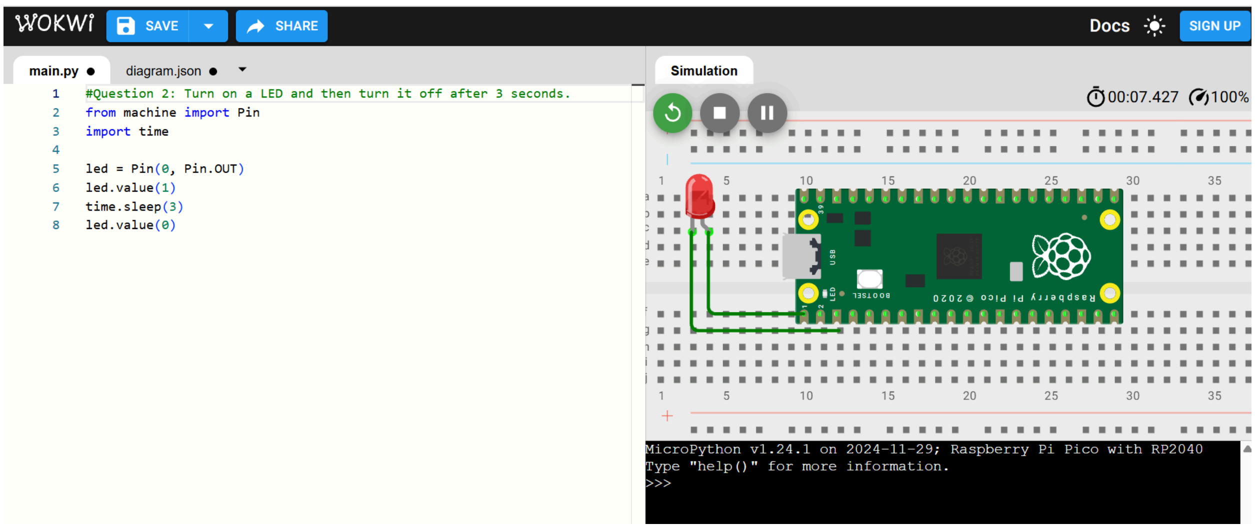


2. **Turn on a LED and then turn it off after 3 seconds.**

   Code

```
from machine import Pin
import time

led = Pin(0, Pin.OUT)
led.value(1)
time.sleep(3)
led.value(0)
```

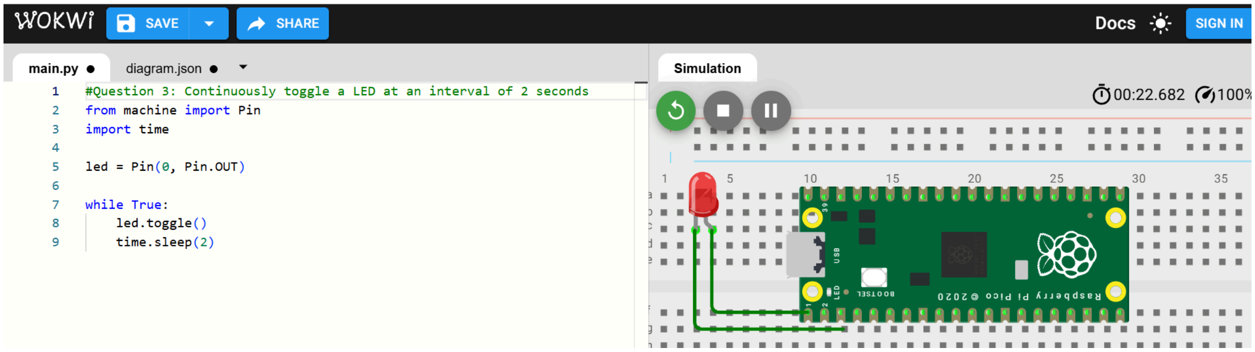


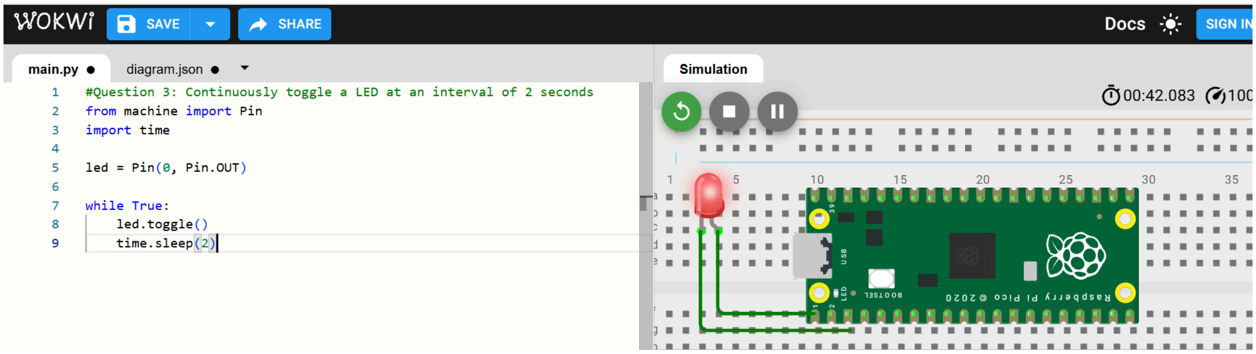


3. **Continuously toggle a LED at an interval of 2 seconds**

Code

```
from machine import Pin
import time

led = Pin(0, Pin.OUT)

while True:
    led.toggle()
    time.sleep(2)
```



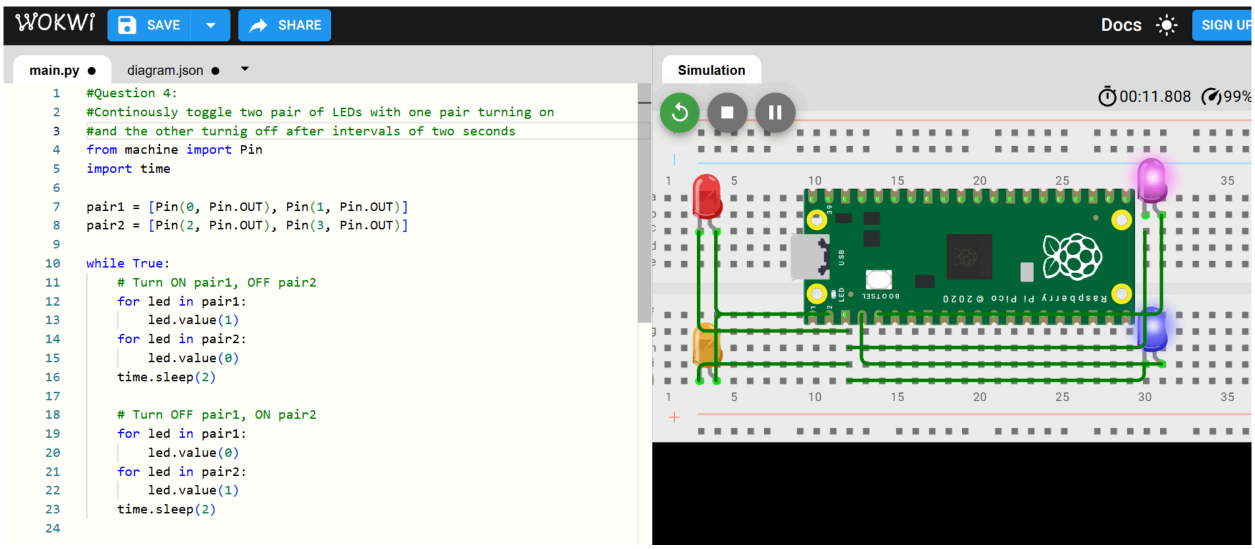


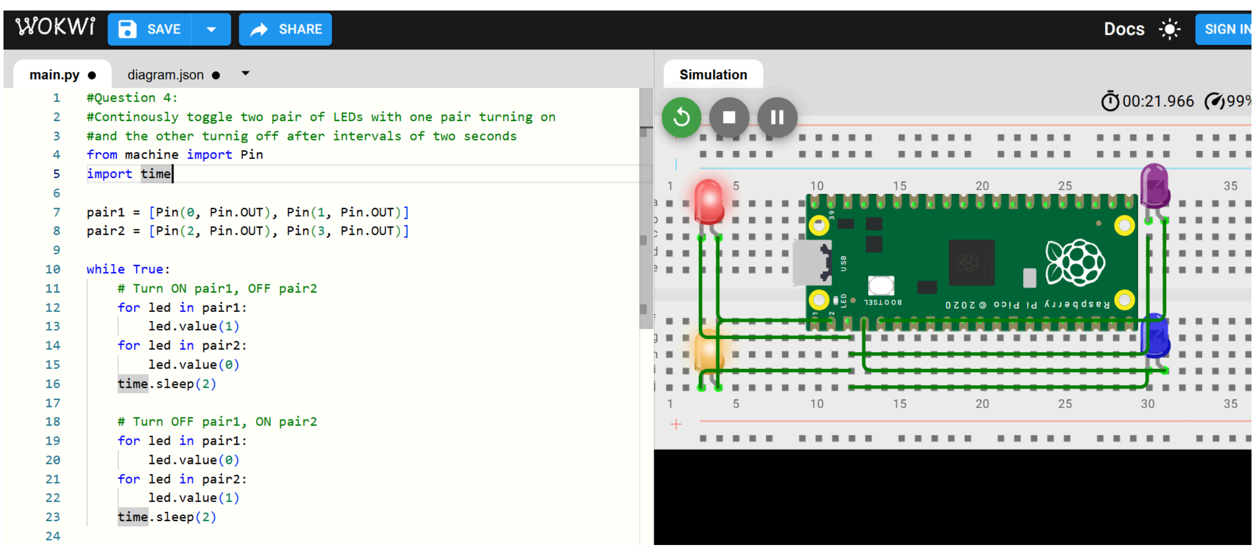
4. **Continously toggle two pair of LEDs with one pair turning on and the other turnig off after intervals of two seconds**

Code

```
#Question 4:
#Continously toggle two pair of LEDs with one pair turning on
#and the other turnig off after intervals of two seconds
from machine import Pin
import time

pair1 = [Pin(0, Pin.OUT), Pin(1, Pin.OUT)]
pair2 = [Pin(2, Pin.OUT), Pin(3, Pin.OUT)]

while True:
    # Turn ON pair1, OFF pair2
    for led in pair1:
        led.value(1)
    for led in pair2:
        led.value(0)
```

```
time.sleep(2)

# Turn OFF pair1, ON pair2
for led in pair1:
    led.value(0)
for led in pair2:
    led.value(1)
time.sleep(2)
```

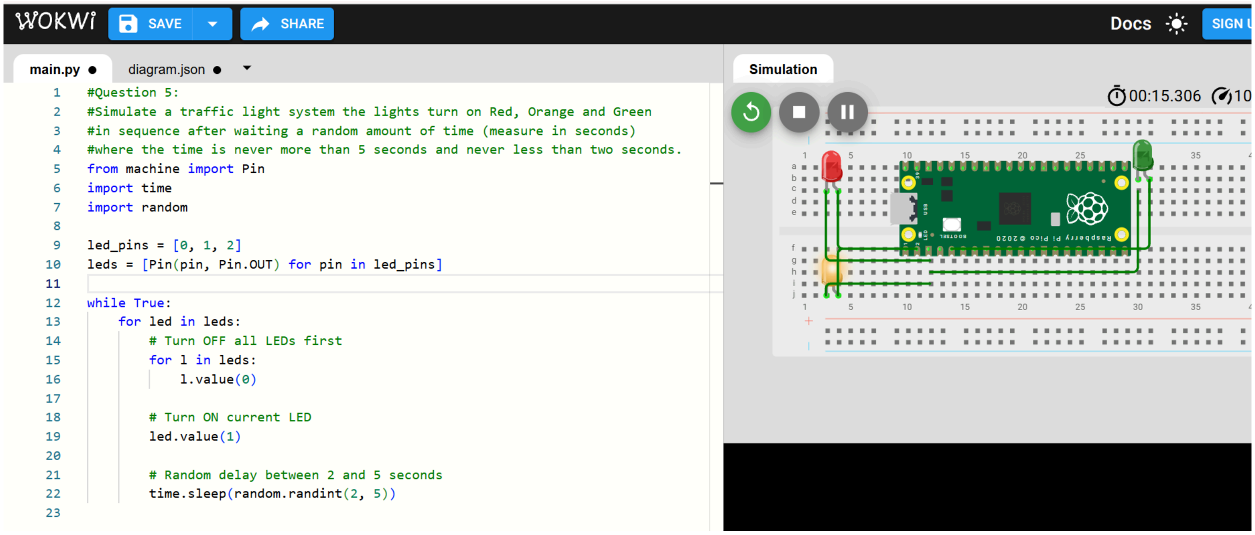


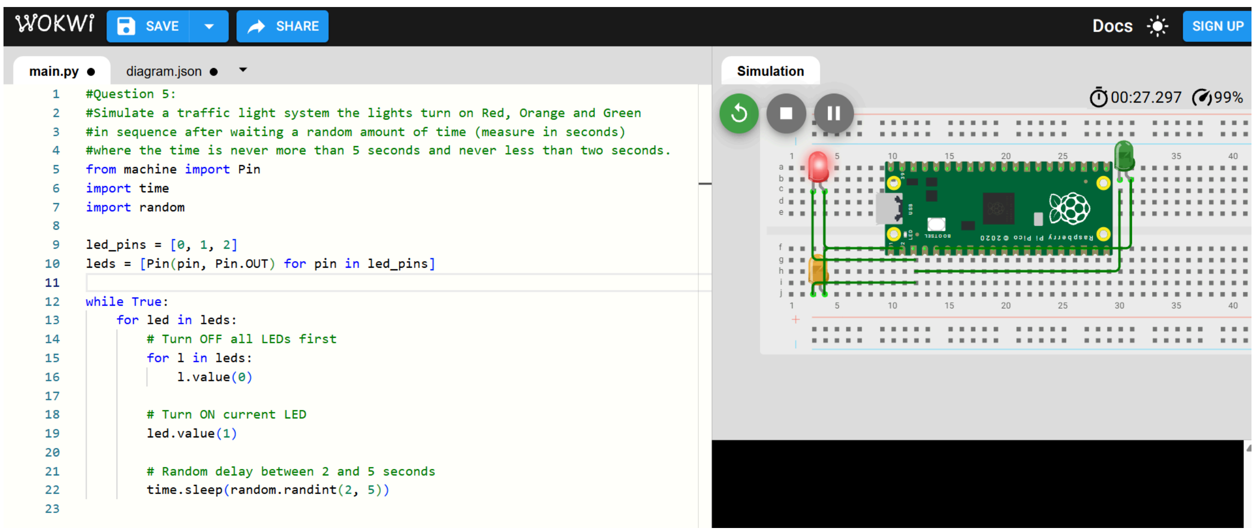


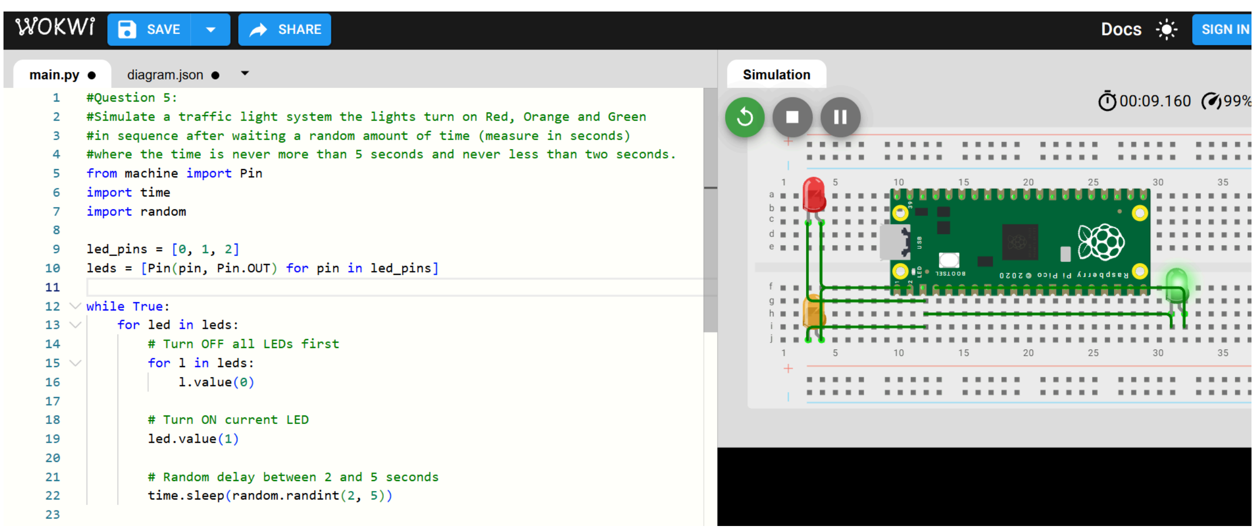
5. **Simulate a traffic light system the lights turn on Red, Orange and Green in sequence after waiting a random amount of time (measure in seconds) where the time is never more than 5 seconds and never less than two seconds.**

Code

```
#Question 4:
from machine import Pin
import time
import random

led_pins = [0, 1, 2]
leds = [Pin(pin, Pin.OUT) for pin in led_pins]

while True:
    for led in leds:
        # Turn OFF all LEDs first
        for l in leds:
            l.value(0)

        # Turn ON current LED
        led.value(1)

        # Random delay between 2 and 5 seconds
        time.sleep(random.randint(2, 5))
```


WOKWI
SAVE
SHARE
Docs
SIGN U
main.py
diagram.json
Simulation
00:15.306
10
#Question 5:
#Simulate a traffic light system the lights turn on Red, Orange and Green
#in sequence after waiting a random amount of time (measure in seconds)
#where the time is never more than 5 seconds and never less than two seconds.
from machine import Pin
import time
import random
led_pins = [0, 1, 2]
leds = [Pin(pin, Pin.OUT) for pin in led_pins]
while True:
for led in leds:
# Turn OFF all LEDs first
for l in leds:
l.value(0)
# Turn ON current LED
led.value(1)
# Random delay between 2 and 5 seconds
time.sleep(random.randint(2, 5))

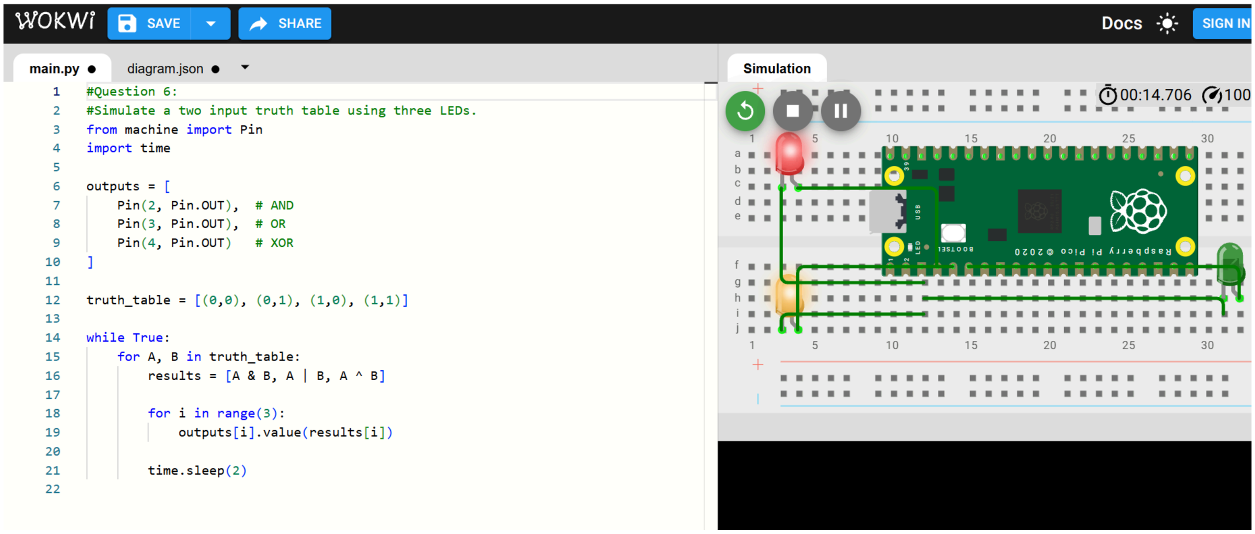

WOKWI
SAVE
SHARE
Docs
SIGN UP
main.py
diagram.json
Simulation
00:27.297
99%
#Question 5:
#Simulate a traffic light system the lights turn on Red, Orange and Green
#in sequence after waiting a random amount of time (measure in seconds)
#where the time is never more than 5 seconds and never less than two seconds.
from machine import Pin
import time
import random
led_pins = [0, 1, 2]
leds = [Pin(pin, Pin.OUT) for pin in led_pins]
while True:
for led in leds:
# Turn OFF all LEDs first
for l in leds:
l.value(0)
# Turn ON current LED
led.value(1)
# Random delay between 2 and 5 seconds
time.sleep(random.randint(2, 5))

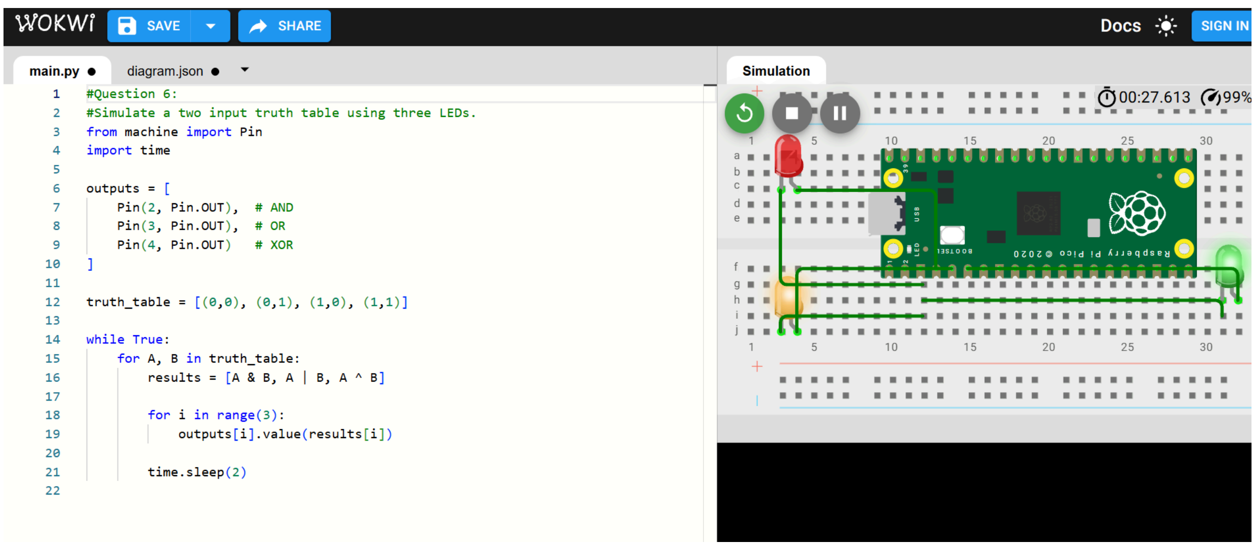

WOKWI
SAVE
SHARE
Docs
SIGN IN
main.py
diagram.json
Simulation
00:09.160
99%
#Question 5:
#Simulate a traffic light system the lights turn on Red, Orange and Green
#in sequence after waiting a random amount of time (measure in seconds)
#where the time is never more than 5 seconds and never less than two seconds.
from machine import Pin
import time
import random
led_pins = [0, 1, 2]
leds = [Pin(pin, Pin.OUT) for pin in led_pins]
while True:
for led in leds:
# Turn OFF all LEDs first
for l in leds:
l.value(0)
# Turn ON current LED
led.value(1)
# Random delay between 2 and 5 seconds
time.sleep(random.randint(2, 5))

6. **Simulate a two input truth table using three LEDs.**

Code

```
#Question 6:
#Simulate a two input truth table using three LEDs.
from machine import Pin
import time

outputs = [
    Pin(2, Pin.OUT),  # AND
    Pin(3, Pin.OUT),  # OR
    Pin(4, Pin.OUT)   # XOR
]

truth_table = [(0,0), (0,1), (1,0), (1,1)]

while True:
    for A, B in truth_table:
        results = [A & B, A | B, A ^ B]

        for i in range(3):
            outputs[i].value(results[i])

        time.sleep(2)
```





7. **Create a system of eight LEDs arranged in a circular fashion. Create a simulation by randomly turning on the LEDs in sequence. After completion of a complete round decrease the time between the LEDs turning on by an amount delta.**

Code

```
#Question 7:
from machine import Pin
import time
import random


leds = [Pin(i, Pin.OUT) for i in range(8)]
```

```
while True:
    random.shuffle(leds)

    for led in leds:
        led.value(1)
        time.sleep(0.3)
        led.value(0)
```

8. **Create a square formation using LEDs of size 4 x 4. Turn on the LEDs individually passing row by row from top to bottom and then in the reverse direction continously.**